

# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE

**SOFTWARE ENGINEER – HCL Technologies | Chennai, India** **09/2021 – 01/2025 (3.5 Years)**

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- Developed and integrated systems for **NPC AI** and **general gameplay** mechanics using C++ and Unreal Engine, enhancing player interaction and contributing to AAA game features.
- Implemented **high-performance code** for AAA game systems, achieving **99.9% stability** under load across PC and console platforms through rigorous testing and debugging.
- Constructed **UI systems** for game interfaces, ensuring seamless user experience and contributing to the successful deployment of features on multiple target platforms.
- Managed **source control** via Git and utilized bug reporting software (Jira) to track and resolve issues, improving team workflow efficiency by **15%**.
- Provided direct **engineering support** to art and design disciplines, translating technical requirements into functional game systems and resolving cross-functional blockers.
- Refactored existing **Python scripts** to automate data processing tasks, reducing manual effort by **20 hours per week** and improving data pipeline reliability.

## SKILLS

**Programming Languages** – C++, Python, Java, SQL, R, DAX, Groovy

**Game Engines & Tools** – Unreal Engine, Jira, Confluence, Git, Jenkins, Splunk, Dynatrace

**Core Engineering** – AI Systems, Gameplay Systems, UI Development, 3D Mathematics, Profiling, Debugging, REST APIs

**Cloud & Infrastructure** – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux

**Data Analytics & BI** – Power BI, Tableau, Pandas, NumPy, Scikit-learn, Data Modelling, Data Warehousing

**Methodologies** – Agile, Scrum, Change Management, Test-Driven Development

## EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin** **02/2026**

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**BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India** **01/2021**

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## PROJECTS

**AI-Driven NPC Behavior System for AAA Open-World Game**

- Engineered a complex **NPC AI system** using C++ and Unreal Engine's Behavior Trees, enabling dynamic decision-making for **over 100 distinct NPC types** in a large-scale open-world environment.
- Developed **gameplay systems** for NPC pathfinding and environmental interaction, reducing navigation errors by **30%** and improving immersion for players on PC and consoles.

- Integrated **profiling and debugging tools** to optimize AI performance, achieving an average frame rate increase of **10 FPS** during intense NPC-heavy scenes.

#### **Automated Transaction Monitoring Dashboard (HCL Technologies)**

- Developed a **payments operations dashboard** to monitor transaction flows and operational alerts for a financial-services client, processing over **1 million transactions daily**.
- Implemented **Python scripts** for ETL processes, consolidating data from multiple sources into a PostgreSQL database for real-time reporting and transaction monitoring.
- Provided **production support** for the dashboard, ensuring 99.9% uptime and responding to P1/P2 incidents within SLA targets.

### **ACHIEVEMENTS & CERTIFICATIONS**

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)